

Sadeeq Adeyemi

Computer Vision & Perception Engineer

Multi-Object Tracking · Sensor Fusion · State Estimation · Deep Learning for Robotics & Autonomous Systems

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PROFILE

Perception engineer working on one problem: **keeping track of what a robot cannot see**. I build multi-object trackers that reason explicitly about occlusion, fuse cameras, lidar and radar, and test every claim with controlled ablations over many random seeds, reported with confidence intervals. A civil and structural engineering background (fabrication, field testing, defect detection in steel) shapes how I work: measure, cross-check against a reference, and state the limits.

RESEARCH

Occlusion-Aware Multi-Object Tracking for Lunar Surface Operations

2026 · independent research

Python · PyTorch3D · JavaScript / Three.js · MATLAB — github.com/sadeeqgandalf/lunar-occlusion-tracking

Can a rover's mast camera keep astronauts' identities when they walk behind boulders or into long south-pole shadows (Artemis EVA scenario)?

- **Visibility model.** 3-D line of sight from a 2.2 m mast camera to 16 points on each body against half-buried ellipsoid boulders, plus cast shadows from a low polar sun, giving each track an expected detection probability.
- **Tracker.** Constant-velocity EKF per track, χ^2 gating, Hungarian assignment with an explicit missed-detection option, Bernoulli track existence. Occlusion-aware variants treat a miss as evidence of absence only in proportion to expected visibility; a negative-information update, $p(x \mid \text{miss}) \propto p(x)(1 - P_D(x))$, keeps hidden estimates inside the blind zone.
- **Results** (20 seeds $\times$ 300 s, identical detections, Wilson 95% intervals). Identities kept through 1–5 s occlusions: **0% naive $\rightarrow$ 46% with negative information $\rightarrow$ 89% with appearance re-ID**. ID switches per run: 75.6 $\rightarrow$ 23.3 with one camera, **6.6** fusing camera, lidar and a second (lander) camera.
- **Sensor-fusion finding.** Adding sensors to a tracker without a visibility model raises ID switches 3.4 $\times$ (75.6 $\rightarrow$ 254): every sensor that cannot see a person reports them gone. A second viewpoint helped more than a better sensor on the same mast.
- **PyTorch3D rendered-camera pipeline.** Built PyTorch3D from source on Apple Silicon (C++20 patch). Rendered the scene from the rover mast and a lander camera; detections come from pixels: visible fraction from silhouettes, range from the depth map, bearing from pixel position, and a colour appearance signature. Static scenery rendered once and per-person windows composited by depth: exact, $\sim 20\times$ faster (0.13 s/frame).
- **Re-identification and PHD.** DeepSORT-style appearance likelihoods and long-gap re-ID raise same-ID after hides over 5 s from 8% to 50%; an identical-suit control (28%) bounds what appearance can contribute. A particle PHD filter with visibility-dependent P_D maps where hidden people can be.
- **Verification.** Python port reproduces the JavaScript engine exactly (cross-check test); MATLAB port compared against MathWorks *trackerGNN*, *trackerJPDA* (JIPDA) and *trackerTOMHT*. Metrics: CLEAR-MOT, IDF1, GOSPA, occlusion survival by duration. Every table regenerates from one script, stamped with its git commit.

Closed-Loop Autonomous Driving Stack

2026

Python · SAM 3 (Hugging Face transformers) · sensor fusion · planning and control

- Full stack that sees only sensor data, never simulator ground truth: EKF localization (IMU, wheel odometry, GNSS, lane-marking map matching); open-vocabulary camera segmentation with SAM 3, lifted to 3-D on lidar; radar Doppler; class-aware CTRV/CV multi-object tracker; prediction; Frenet lattice planner; LQR lateral and PID longitudinal control; independent safety monitor with corroborated-track emergency braking.
- Occlusion reasoning spawns "phantom pedestrians" in blind zones. Monte Carlo over 10 scenarios $\times$ 10 seeds: **100/100 runs pass** with 0 collisions. Ablation: without occlusion reasoning, a pedestrian stepping out from behind a bus is struck in **5 of 10** runs.

Open-Vocabulary Segmentation and Tracking on Driving Video 2026

SAM 3 · KITTI tracking · BDD100K · TrackEval — github.com/sadeeqgandalf/ComputerVision_Research

- Text-prompted segmentation and video tracking (MOTS) with Meta's SAM 3; per-prompt masklets merged into one global identity space; evaluation with TrackEval (HOTA).

Sensor Fusion from Raw Signals 2026

MATLAB · Sensor Fusion and Tracking, Automated Driving and Signal Processing toolboxes

- FMCW radar from raw IF samples to range-Doppler maps, CFAR detection, angle estimation and calibrated detections with ego-motion; accuracy compared with the Cramér–Rao bound. IMU noise by Allan variance and GNSS error models. Algorithms written from scratch and cross-validated against MathWorks reference implementations, with unit tests on every numerical claim.

Lidar Perception and Moving-Object Segmentation on Argoverse 2 2026

Python · Open3D · scikit-learn · Rerun

- Ground removal, DBSCAN clustering and PCA-oriented bounding boxes on real sweeps. Moving-object segmentation from SE(3) ego-motion compensation, nearest-neighbour motion residuals and local scene-flow coherence, graded against ground-truth movers in the city frame.

Navigation Filters under Faults 2026

JavaScript · MATLAB — Mars rover, aircraft, and spacecraft docking with NASA's Gateway

- Dead reckoning, EKF, UKF and particle filter on identical sensor streams under injected faults, following *Probabilistic Robotics*; consistency checked with NEES/NIS, every analytic Jacobian verified by finite differences. Clohessy–Wiltshire rendezvous with a 6-state EKF and gated approach (MATLAB port).

Deep Learning for Vision 2025 – 2026

- **Vision Transformer, from scratch (PyTorch).** 7-class dermoscopy (HAM10000), compared against a pretrained ViT; class-weighted loss, warm-up schedule, per-class AUC, [CLS] attention maps.
- **NeRF reimplementations (in progress).** Ray generation, positional encoding, hierarchical sampling, volume rendering; 22 tests.
- **Dashcam perception.** YOLOv8 detection, IoU tracking with coasting, lane detection, distance and time-to-collision; 24 regression tests.

EXPERIENCE

Robotics Research Assistant 2025 – present · Orono, Maine

University of Maine, Robotics Laboratory

- Build and integrate hardware and software for mobile robot platforms: sensor mounting, wiring, and ROS 2 nodes for autonomous navigation.
- Implement EKF state estimation fusing IMU, wheel odometry and lidar for indoor and outdoor localization.
- Develop costmap-based path planning and obstacle avoidance with global and local planners in ROS 2; build point-cloud processing and detection pipelines for autonomous-vehicle research.

Materials Engineer 2024 · CPT placement, United States

R.W. Gillespie

- Field and laboratory materials testing (compaction, concrete strength, aggregate gradation) to ASTM standards on active construction sites, with zero non-conformance incidents across inspected batches.

NewTech Steel

- Designed and deployed an in-house computer vision model for real-time detection of fatigue and structural defects in steel members, integrated into production monitoring.
- Built an automated testing framework that cut testing time by 50%; maintained production models with versioning and performance monitoring.
- Led fabrication and erection of structural steel for a large factory, a church auditorium and hostel roofing, coordinating with design engineers on connection detailing, assembly sequencing and dimensional quality control.

Earlier: Industrial Engineering Trainee, Federal Capital Development Authority, Nigeria (2017–2018), sewage and roadway infrastructure; Engineering Intern, Kokun Integrated Services, Nigeria (2016–2017).

EDUCATION

MS, Data Science & Engineering <i>University of Maine</i>	2025 – present	MS, Structural Engineering <i>University of Maine</i>	2023 – 2025
MSc, Project Management <i>University of Portsmouth, UK</i>	2023	BEng, Civil Engineering <i>Federal University of Technology, Akure, Nigeria</i>	2019

SKILLS

Tracking & estimation	KF / EKF / UKF, particle and PHD filters, GNN, JPDA, MHT, Bernoulli existence, negative information, NEES/NIS; CLEAR-MOT, IDF1, HOTA, GOSPA
Vision & deep learning	PyTorch, PyTorch3D, SAM 3, YOLOv8, Faster R-CNN, Vision Transformers, NeRF, appearance re-ID, Hugging Face
3-D & sensors	Camera geometry, depth-buffer visibility, lidar point clouds (Open3D), FMCW radar DSP and CFAR, IMU/GNSS noise modelling
Robotics	ROS 2, colcon, costmap navigation, closed-loop planning and control (Frenet, LQR, PID)
Engineering	Python, C++, MATLAB, JavaScript / Three.js, SQL, Git, Docker, AWS, FastAPI

CERTIFICATIONS

AWS Certified Machine Learning – Specialty · AWS Certified Solutions Architect – Associate · PyTorch Deep Learning Specialization (DeepLearning.AI)